

CHEN SHU YAN



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Summary

Third-year Software Engineering student at UTM and former Robocon Vice President with a focus on high-performance system integration and AI application development. Experienced in bridging native mobile platforms with robotics middleware using C++, JNI, and FastDDS. Currently developing an AI-powered Insurance Intelligence system utilizing RAG (Retrieval-Augmented Generation) and vector databases. Seeking a 20-week internship **from Sept 21, 2026 – Feb 5, 2027** to apply my technical background in full-stack engineering and AI optimization.

Projects

AI-Powered Insurance Intelligence (FYP & Competition)

March 2026 to Present

- **Architected a full-stack prototype using Django, Next.js, and Supabase**, featuring end-to-end PDF ingestion and structured policy data extraction.
- **Developed an "Emergency RAG" pipeline utilizing all-MiniLM embeddings and Supabase pgvector** to deliver fact-grounded insurance guidance with precise source references.
- **Implemented core optimization features**, including premium overlap/gap detection and a "Family Vault" module with **RBAC-style policy** sharing secured by Supabase-authenticated APIs.

Robot Control & Map Visualization App (Andriod)

January 2025 - May 2025

- **Developed a native Android app** in Kotlin to visualize live robot telemetry, bridging the frontend to a C++ backend via a custom JNI (Java Native Interface).
- **Optimized data throughput** by tuning FastDDS Quality of Service (QoS) settings, ensuring stable telemetry updates and reducing data lag under high network congestion.
- **Troubleshooted system-level data mismatches** by identifying coordinate frame transformation errors, ensuring accurate real-time position tracking for competition operators.

PawSure Pet Care Superapp (AI Integration)

Oct 2025 - Feb 2026

- **Deployed an AI health scanning module** using **ONNX Runtime**, achieving an **88% precision rate** in detecting early physical symptoms via mobile camera feeds.
- **Developed a modular backend** in **NestJS** to handle large-scale image uploads and metadata logging for iterative model retraining cycles.

Extra - Curricular

Vice President (UTM Robocon Team) | July 2024 - Jun 2025

- **Technical Leadership:** Co-led a team of 35 members to develop autonomous robots for national competitions, ensuring seamless integration between software, electronic and mechanical
- **System Architecture:** Developed and led the development of an Android-based monitoring system using C++ and FastDDS, improving real-time data visibility for operators during trials.

State Athlete (Volleyball) | SUKMA Johor | 2024

- Represented Johor State in SUKMA (National Youth Games), demonstrated elite-level discipline and teamwork by balancing rigorous national-level training with Dean's List
- Developed high-pressure composure and strategic coordination through competitive play, skills directly transferable to managing complex, real-time technical deployments.

Education

Universiti Teknologi Malaysia

Bachelor of Computer Science
(Software engineering)

CGPA: 3.82

2023 - 2027 (Expected)

Kolej Matrikulasi Pahang

Physcial Science

CGPA: 4.00

2022 - 2023

Skills

- **Programming Languages:** C++, Java, JavaScript, Kotlin, SQL, Python
- **AI & Machine Learning:** RAG (Retrieval-Augmented Generation), Vector Databases (pgvector), Embeddings (MiniLM), ONNX Runtime, YOLO (Exposure)
- **Software & Frameworks:** Flutter, Next.js, Supabase, NestJS, Django, React, PostgreSQL
- **Middleware & Systems:** FastDDS, JNI (Native Development), ROS 2, Linux/Ubuntu

Language

- English
- Chinese
- Malay